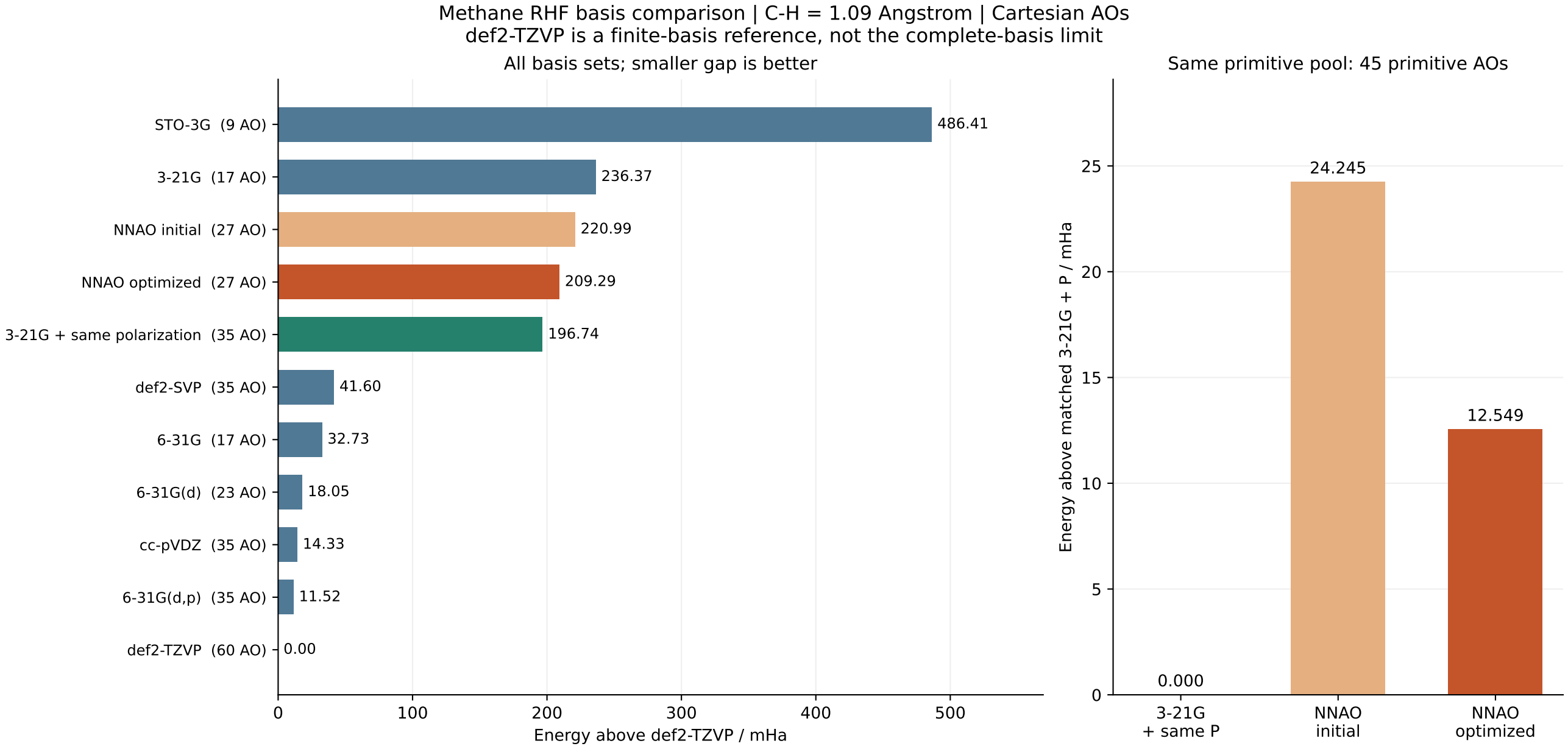
Methane RHF basis comparison | C-H = 1.09 Angstrom | Cartesian AOs
def2-TZVP is a finite-basis reference, not the complete-basis limit
All basis sets; smaller gap is better
STO-3G (9 AO)
3-21G (17 AO)
NNAO initial (27 AO)
NNAO optimized (27 AO)
3-21G + same polarization (35 AO)
def2-SVP (35 AO)
6-31G (17 AO)
6-31G(d) (23 AO)
cc-pVDZ (35 AO)
6-31G(d,p) (35 AO)
def2-TZVP (60 AO)
486.41
236.37
220.99
209.29
196.74
41.60
32.73
18.05
14.33
11.52
0.00
0
100
200
300
400
500
Energy above def2-TZVP / mHa
Same primitive pool: 45 primitive AOs
Energy above matched 3-21G + P / mHa
0
5
10
15
20
25
24.245
12.549
0.000
3-21G
+ same P
NNAO
initial
NNAO
optimized